

# MAINTENANCE-AI

## The Paradigm Shift in Smart Property Operations & Maintenance Systems

**Author:** Adeer International Engineering & Products Group

**Target Audience:** Property Managers, Asset Owners, Operations Directors

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**Classification:** Confidential / Proprietary

# 1. Problem Statement

Property management companies face significant challenges in handling maintenance requests:

- **Manual Coordination:** Requests come in via phone, email, or in-person, requiring property managers to manually classify, prioritize, and dispatch service providers.
- **Delayed Response Times:** Without automated triage, critical issues (e.g., gas leaks, flooding) may sit in the same queue as minor issues (e.g., a loose doorknob).
- **Poor Visibility:** Tenants lack real-time visibility into the status of their requests, leading to frustration and repeat contacts.
- **High Overhead:** Property managers spend excessive time on administrative tasks instead of strategic decision-making.

# 2. The MaintenanceAI Solution

**MaintenanceAI** is an AI-powered maintenance request management system that automates the entire lifecycle of a maintenance request — from submission to resolution. The system uses Google's Gemini AI to analyze natural language descriptions from tenants and automatically:

- **Classify:** categorizes the issue into one of 10 types (Plumbing, Electrical, HVAC, Structural, etc.).
- **Assess Severity:** calculates risk level dynamically (Low, Medium, High, Critical).
- **Assign Priority:** sorts urgency level from 1 (highest) to 5 (lowest).
- **Generate Remediation Plans:** provides interactive technical resolution checklists.
- **Predictive Cost:** estimates expected repair cost range for budget approvals.
- **Automated Routing:** immediately assigns the ticket to targeted service providers.

# 3. Target Users & System Benefits

| Role              | How They Use MaintenanceAI                                                                                                                 |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Tenants           | Submit maintenance requests in plain language. Receive instant confirmation with AI-generated ticket details and real-time status updates. |
| Property Managers | Use the Operations Dashboard to view, filter, assign, and manage all requests. Monitor critical issues with priority-based views.          |
| Service Providers | Receive automated assignment notifications with issue details, severity, and recommended action steps.                                     |

## **4. Key Features & Integrations**

### **4.1 Natural Language Request Submission & Multi-Modal Support**

Tenants describe their issue in their own words and can upload images. The AI handles the rest, analyzing both text and attached visual evidence to get accurate context.

### **4.2 AI-Powered Classification & Prioritization**

Using Google Gemini, the system extracts structured data from unstructured text and images, filling in the category, severity level, priority metrics, action steps, and cost estimate dynamically.

### **4.3 Operations Dashboard & Invoice Generation**

A comprehensive, real-time dashboard for property managers featuring summary statistics cards, search and multi-filtering, detailed request logs, automated PDF invoice generation, and one-click status updates.

### **4.4 Automated Communication & Ticket Updates**

The system automatically tracks communication between tenants and admins. When comments are posted, the system preserves history logs and automatically triggers Gemini to regenerate issue summaries.

## 5. AI Capabilities & Architecture Specifications

| Capability                 | Description                                                                    |
|----------------------------|--------------------------------------------------------------------------------|
| Multi-Modal Analysis       | Processes free-form text and images to extract structured maintenance data.    |
| Multi-label Classification | Simultaneously determines category, severity, and priority.                    |
| Summarization              | Converts informal language and history logs into professional issue summaries. |
| Action Recommendation      | Generates step-by-step resolution plans.                                       |
| Cost Estimation            | Provides estimated repair cost ranges for billing.                             |

## 6. Technology Stack

| Layer               | Technology                                                       |
|---------------------|------------------------------------------------------------------|
| Frontend            | Next.js 16 (App Router), React, Tailwind CSS                     |
| Backend             | Next.js API Routes (server-side)                                 |
| Database            | SQLite via Prisma ORM                                            |
| AI Engine           | Google Gemini 2.0 Flash                                          |
| Deployment          | Fly.io (Containerized Docker deployment with persistent volumes) |
| Document Generation | Python (ReportLab) & jsPDF                                       |

## 7. Expected Business Impact

- **95% faster classification:** AI processes requests in seconds vs. minutes of manual review.
- **60% cost reduction:** Reduced administrative overhead and faster issue resolution.
- **24/7 availability:** Tenants can submit requests anytime, AI classifies immediately.
- **Improved tenant satisfaction:** Real-time visibility and faster response times.
- **Data-driven decisions:** Dashboard analytics help managers identify recurring issues.

## 8. Future Enhancements

- **Predictive Maintenance:** Use historical data to predict and prevent common issues.
- **Multi-language Support:** Leverage AI for automatic translation of requests.

- **Integration with IoT:** Connect smart building sensors for automatic issue detection.
- **Mobile App:** React Native companion app for tenants and service providers.